

CHIRAG APARADH

Software Engineer | Backend | Distributed Systems | AI

chiragaparadh@gmail.com | Mumbai, India | github.com/git-chirag | linkedin.com/in/chirag-aparadh | Portfolio

EXPERIENCE

Software Engineer (Associate Consultant)

Oracle Financial Services Software

Jul 2023 - Aug 2026 | Mumbai, India

- Built and maintained core Java EE microservices powering HDFC Bank's net banking platform for 2M+ daily active users, sustaining 99.99% uptime through highly available clustered deployments.
- Implemented a BioCatch bot-detection workflow for merchant transactions, triggering OTP-based authentication for suspicious activity and strengthening fraud prevention across 150K daily transactions.
- Built event-driven integrations with Apache Kafka to publish acknowledgements to external broker systems after successful fund transfers.
- Modernized a legacy banking workflow by translating UI-driven business logic into validated backend service APIs and standardized data contracts.

Technologies: Java, Java EE, JavaScript, SQL, Oracle Database, Apache Kafka, Microservices, IBM WebSphere

Machine Learning Engineer - Intern

TechCiti

Aug 2021 - Sep 2021 | Remote

- Designed a loan-eligibility product using age, loan amount, credit score, location, and employment data.
- Trained a logistic regression model with 92% test accuracy and built authentication, profile, and prediction modules.

Technologies: Python, NumPy, Pandas, Scikit-learn

EDUCATION

M.S. in Computer Science

University of Massachusetts Amherst

Currently attending | Aug 2026 - May 2028 (Expected) | Amherst, MA

B.Tech in Computer Science and Engineering

Ramrao Adik Institute of Technology

Mumbai University, 2023 | CGPA: 9.51 / 10

ACHIEVEMENTS

- Finalist in the Meta x Hugging Face OpenEnv Hackathon, reaching the national finale among 31,000+ registered teams.
- Ranked 1108 globally in LeetCode Biweekly Contest 145 and earned the Knight badge.
- Earned a 5-Star Gold Badge in Problem Solving on HackerRank.
- Cleared GATE CS 2023 with AIR 2816 among 75,680 candidates.

INTERESTS

Competitive programming, chess, weight training, AI systems

TECHNICAL SKILLS

- Languages: Java, Python, C, JavaScript, SQL
- Backend: Java EE, FastAPI, Django, React, REST APIs, Microservices, Apache Kafka, Redis
- Data and Cloud: Oracle Database, MySQL, MongoDB, Qdrant, AWS, Docker, Kubernetes, ECS, ECR, Jenkins, Linux
- AI and Core CS: PyTorch, CLIP, NumPy, Pandas, Scikit-learn, Distributed Systems, DSA, OOP, Operating Systems, Networks

CODING PROFILES

LeetCode: eLeet_chirag | Rating: 1866

HackerRank: chiragaparadh

PROJECTS

JiraRL - Stateful RL Environment for LLM

Agents

- Built a stateful, OpenEnv-compatible Jira workflow simulator with structured agent actions, isolated episodes, dependency constraints, deterministic task generation, and executable reward functions.
- Generated and validated 3,500 hint-free procedural training decisions with disjoint train, validation, and test seeds, publishing reproducible datasets and model adapters to Hugging Face.
- Fine-tuned Qwen3-0.6B using 4-bit QLoRA, achieving 100% completion across 60 held-out in-distribution workflow episodes, with 500/500 productive transitions and zero invalid actions.
- Implemented environment-backed GRPO with checkpoint recovery, balanced rewards, neutral rollouts, behavioral audits, and automated promotion gates.
- Diagnosed action collapse and zero-gradient RL optimization using raw trajectories, candidate reward variance, entropy, and gradient statistics; currently extending evaluation to out-of-distribution workflows and recovery states.

Technologies: Python, FastAPI, OpenEnv, Qwen3, QLoRA, GRPO, Hugging Face

Multimodal Image Search System

- Built an asynchronous FastAPI and Celery pipeline that generates CLIP embeddings for semantic and visual image retrieval.
- Indexed vectors in Qdrant and coordinated distributed processing through Redis workers and Amazon S3.
- Containerized and deployed the application with Docker, AWS ECR, and Amazon ECS.

Technologies: FastAPI, PyTorch, CLIP, Qdrant, Redis, Celery, AWS, Docker

Supply Chain Management Using Blockchain

- Built a role-aware Ethereum application for transparent product listings, bidding, purchasing, and traceability across the supply chain.
- Implemented 13 smart-contract functions covering producer, distributor, retailer, and customer workflows.

Technologies: Ethereum, Solidity, React, Tailwind

PUBLICATION

- Exploring the Role of Blockchain in Enhancing Supply Chain (IEEE, Sep 2023)